TargetMind:
LLM-Powered Biomedical Retrieval & Insights

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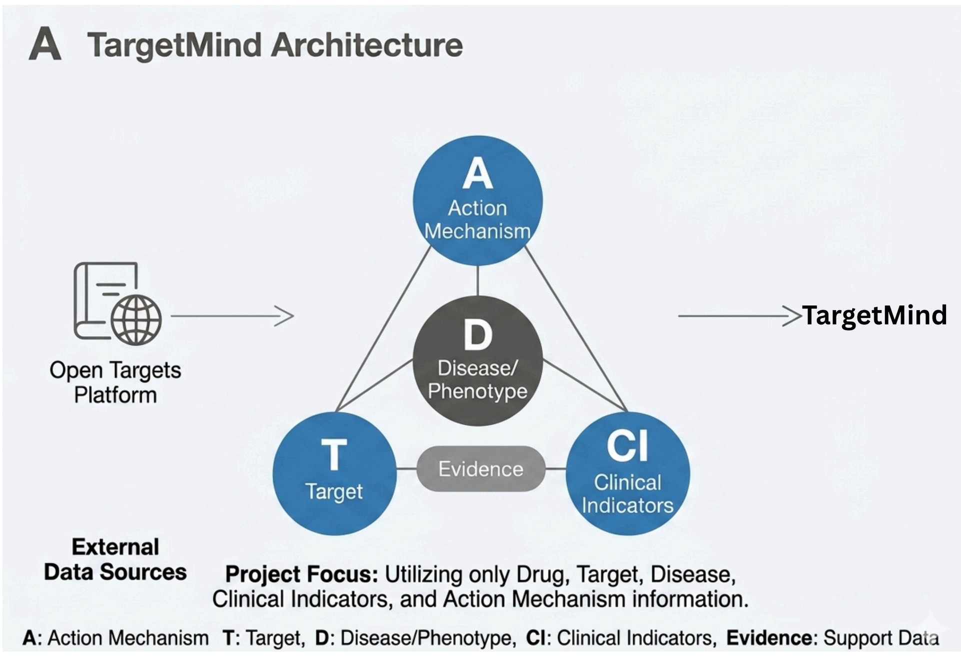
Abstract

TargetMind is a natural language biomedical question-answering system developed to make complex biomedical knowledge more accessible and easier to explore. Built on the Open Targets platform, it enables users to ask questions about genes, diseases, drugs, mechanisms, and biomedical relationships without requiring knowledge of database schemas or structured query languages. The system retrieves evidence-based biomedical information and presents it in a clear, readable format suitable for research exploration and academic use.

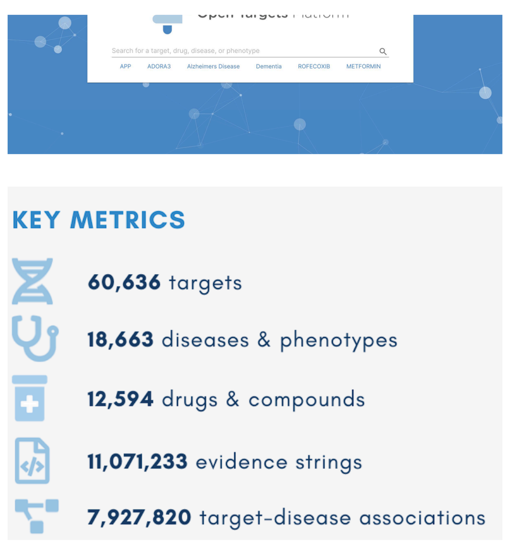


Open Targets

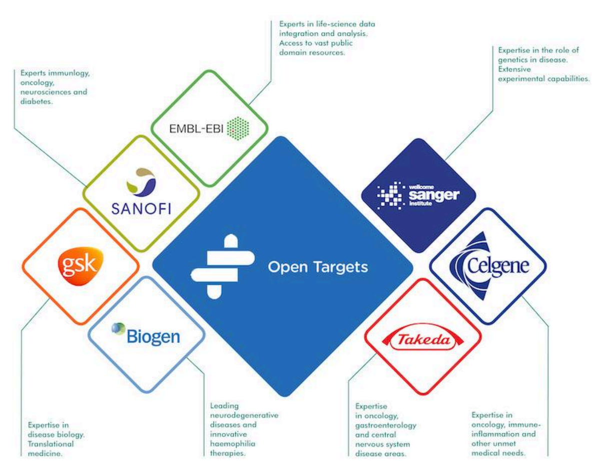
platform.opentargets.org

What is Open Targets?

The Open Targets Platform is a comprehensive, open-access resource designed to facilitate systematic drug target identification and prioritization. It is a collaborative public-private partnership—involving entities such as the European Bioinformatics Institute (EMBL-EBI), Wellcome Sanger Institute, and major pharmaceutical leaders—aimed at bridging the gap between biomedical research and drug discovery.



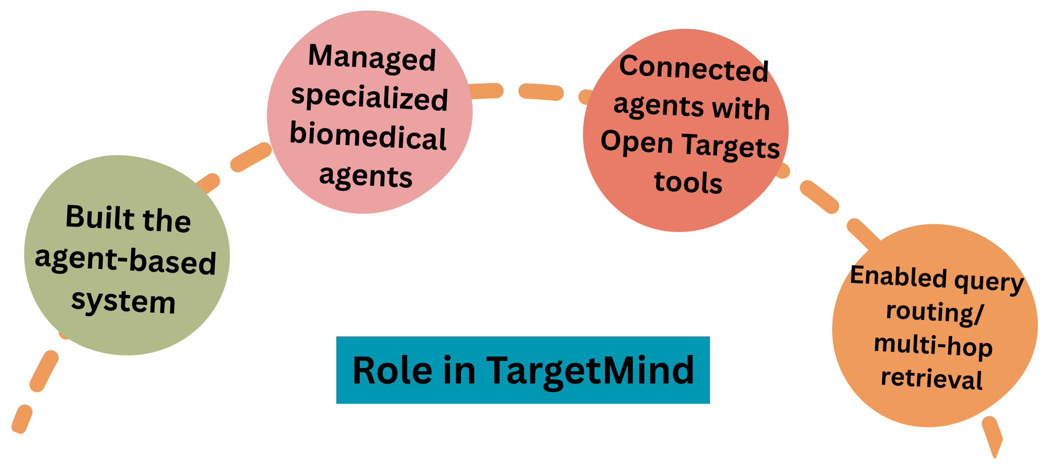
Data sources



PhiData

https://docs.phidata.com

- PhiData is a framework used to build AI agents that can understand tasks, use tools, and generate responses. It supports agent based workflows where multiple agents can work together to complete a task.



Specialist agents

Search Agent
Entity resolution

Orchestrator
Query routing

Target Agent
Gene/target data

Disease Agent
Disease data

Drug Agent
Drug data

Multi-hop agent
Multi-hop reasoning

Methodology

1. Natural Language Query Processing
The system receives a biomedical question in natural language and identifies whether the query is related to a gene/target, disease, drug, mechanism, or relationship between them.

2. Entity Resolution
The main biomedical term in the query is searched in Open Targets and mapped to its correct database identifier, such as an Ensembl ID, EFO ID, or ChEMBL ID.

3. Query Intent Classification
The system determines whether the user query requires a direct lookup or a multi-step biomedical relationship, such as gene-to-disease, disease-to-drug, or target-to-disease-to-drug retrieval. This select the correct path so the system can answer both simple and complex biomedical questions accurately.

4. Structured Tool Selection and API Execution
Based on the resolved entity and query intent, the LLM selects the appropriate predefined tool instead of generating raw queries directly directly. The selected tool contains the required GraphQL query structure and sends the API request to Open Targets.

5. Biomedical Data Retrieval
Relevant structured data is retrieved from Open Targets, including disease associations, drug indications, target relationships, mechanisms of action, and clinical-stage information, association scores, synonym

6. Response Formatting
The returned JSON data is filtered, cleaned, limited to relevant results, and converted into a concise natural language answer for the user interface.

Tools & Tech

Llama 3.3 | Groq | Chainlit | Phidata | Open Target | Python

Tools

Search Tool
search_entity()

Target (Gene) Tools
get_target_diseases()
get_target_drugs()
get_target_synonyms()

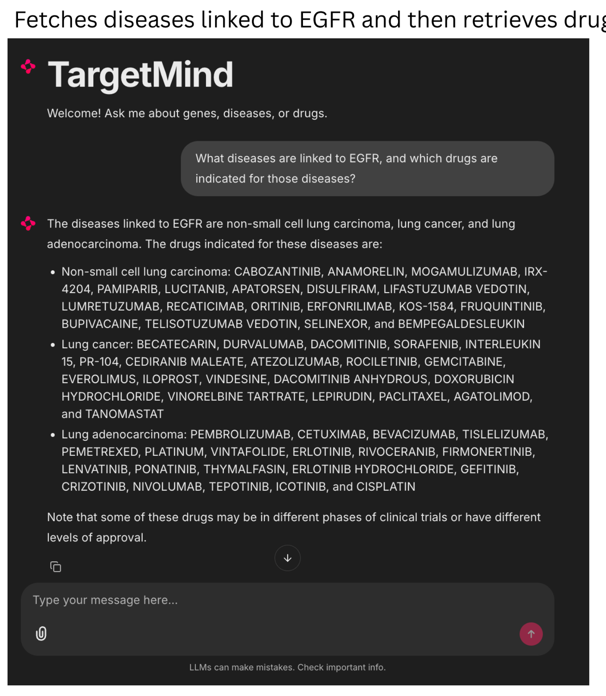
Disease Tools
get_disease_targets()
get_disease_drugs()

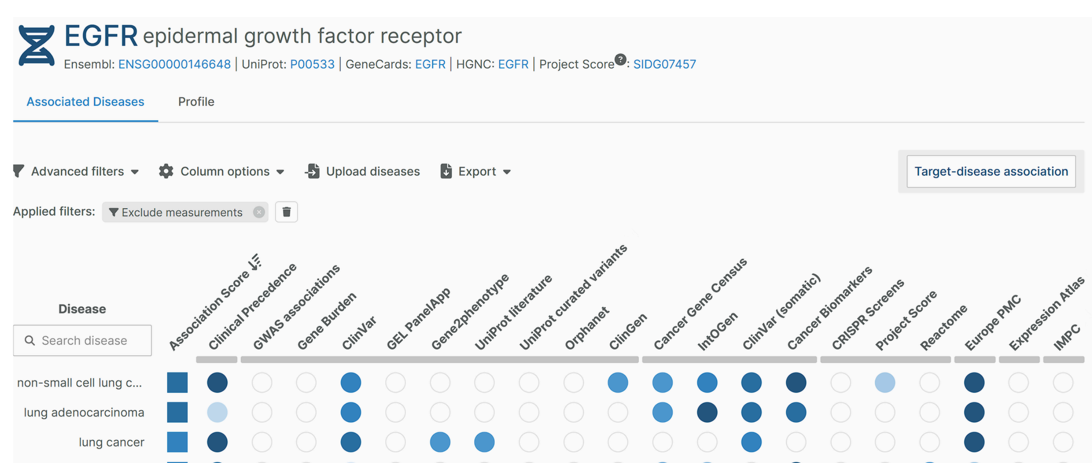
Drug Tools
get_drug_targets()
get_drug_diseases()
get_drug_mechanism()

Multi-hop agent
disease_to_targets_to_drugs()

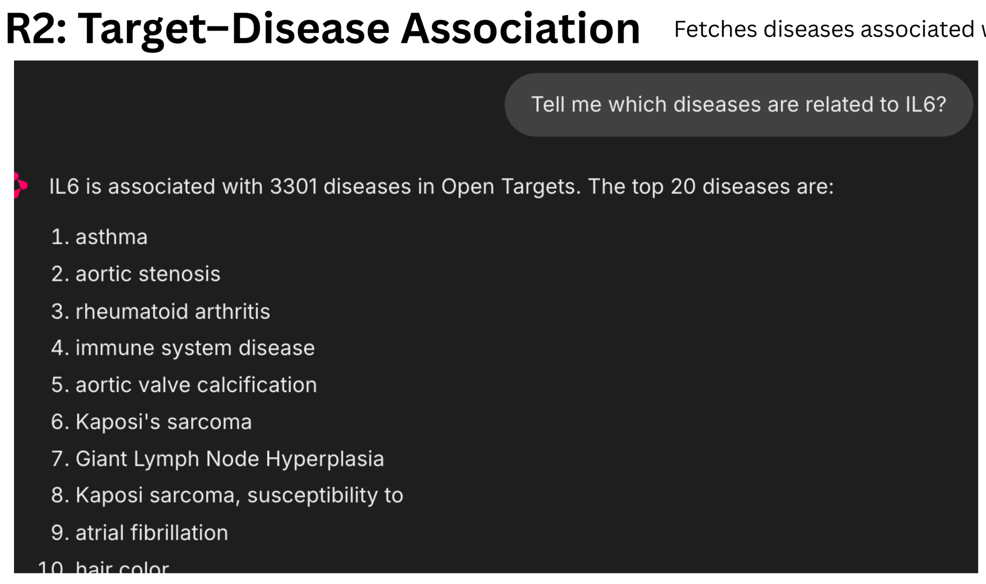
Results

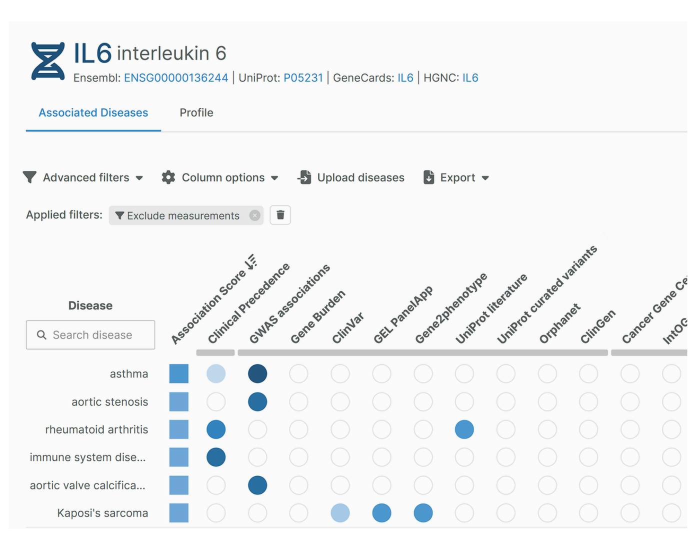
R1: Multi-hop Retrieval
Fetches diseases linked to EGFR and then retrieves drugs indicated for those diseases.



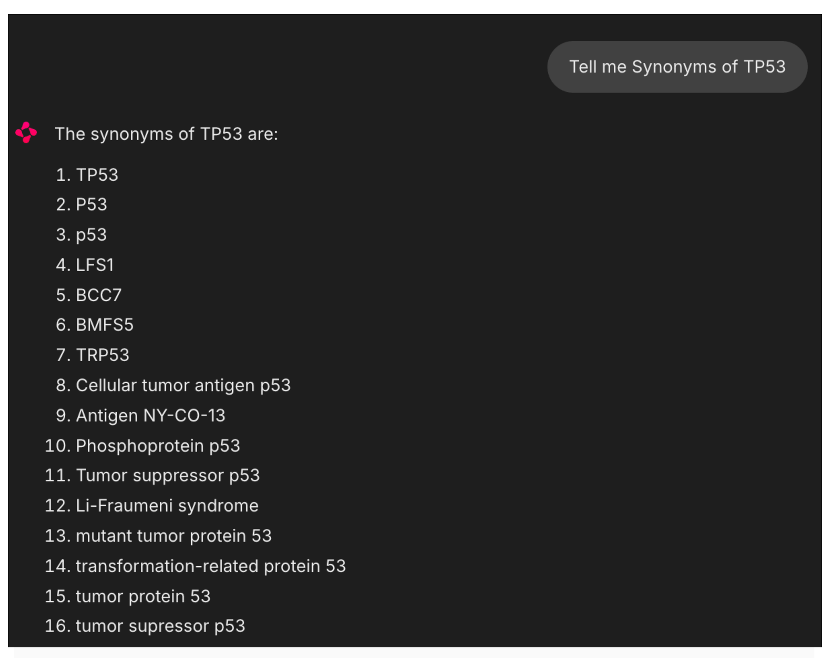


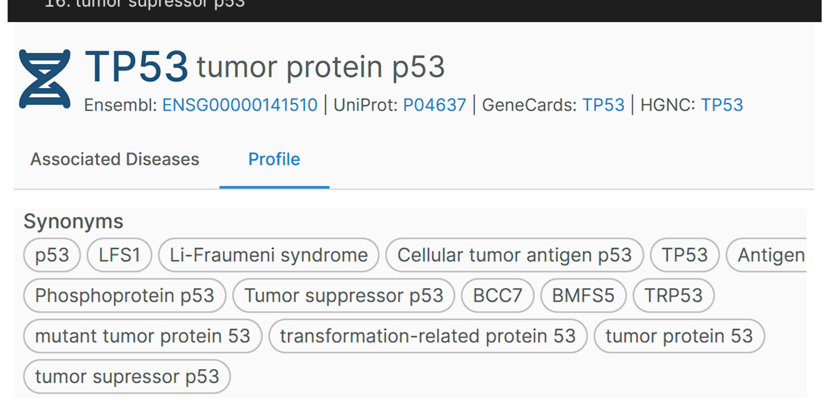
R2: Target–Disease Association Fetches diseases associated with IL6 from Open Targets ranked association data.





R3: Synonym Retrieval & Validation
Fetches known synonyms of TP53 and validates them against the Open Targets profile page.





Conclusion

TargetMind provides an efficient and user friendly way to explore biomedical knowledge from the Open Targets platform. By using a multi-agent LLM architecture, the system can understand natural language queries, identify biomedical entities, retrieve relevant data through GraphQL API calls, and present the results in a clear format. It supports queries related to genes, diseases, drugs, mechanisms of action, and multi-hop biomedical relationships. Overall, the project reduces the complexity of manual database searching and demonstrates how agentic AI can significantly improve biomedical data retrieval and knowledge discovery.

Future Scope

Future work will focus on integrating genomics datasets and biological knowledge graphs. This will enable richer biomedical reasoning, deeper relationship discovery, and broader support for complex queries across genes, diseases, drugs, pathways, and molecular interactions.